

Vincenzo Antonio Isoldi

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Research Interests

My research interests lie at the intersection of geometry, dynamics, and representation theory. My recent work has concerned homogeneous varieties and their secant varieties, tensor geometry, discrete differential geometry, and interactions between algebraic geometry, combinatorics, and game theory. During my PhD, I plan to focus on Riemann surfaces and their moduli, higher Teichmüller theory, and Anosov representations, with an eye toward connecting Lie-theoretic structures with the analytic and dynamical geometry of surfaces.

Education and Training

- 2026– **Ph.D. Student**
Max Planck Institute for Mathematics in the Sciences (MPI MiS)
Supervisor: Prof. Dr. Anna Wienhard.
Research group: Geometry, Groups, and Dynamics.
- 2026 **Pre-Doctoral Fellow**
Max Planck Institute for Mathematics in the Sciences (MPI MiS)
Max Planck Institute of Molecular Cell Biology and Genetics (MPI-CBG)
Supervisors: Profs. İrem Portakal (MPI MiS) and Türkü Özlüm Çelik (MPI-CBG).
Project on algebro-geometric, combinatorial, and game-theoretic aspects in the dynamics of interacting species.
- 2023–2025 **M.Sc. in Mathematics**
University of Trento
Final grade: 110/110 with honors. Final GPA: 30/30.
Thesis: “Schur Apolarity and the Dual Terracini Lemma for Flag Varieties: A Geometric and Representation-Theoretic Approach”, supervised by Prof. Alessandra Bernardi and Dr. Stefano Canino.
- 2020–2023 **B.Sc. in Mathematics**
University of Milan
Final grade: 110/110.
Final seminar: “Plane Algebraic Curves: Local Study and Plücker Formulas”.
- 2019–2020 **B.Sc. in Chemical Engineering**
Polytechnic University of Milan
GPA: 28.75/30. Mathematical coursework GPA: 30/30.
Interrupted to transition to B.Sc. in Mathematics.

Publications and Preprints

1. **Secant varieties of flag varieties via Schur apolarity**, with A. Bernardi and S. Canino. arXiv: [2609.09301](https://arxiv.org/abs/2609.09301).
2. **Stable Coexistence in Ecologies and Games**, with T. O. Çelik, I. Portakal, and G. Zucal. arXiv: [2609.09286](https://arxiv.org/abs/2609.09286)
3. **Neural Point-Forms**, with J. Hume, E. S. Ivshina, K. Maggs, P. Misof, and B. Trentini. arXiv: [2605.15524](https://arxiv.org/abs/2605.15524). Submitted.

Honors, Awards, and Fellowships

- 2026 **Pearcy/Heep Graduate Mathematics Fellowship**, Texas A&M University
Awarded a one-time \$2,500 fellowship in addition to a fully funded PhD offer. Declined.
- 2026 **Pre-Doctoral Fellowship**, Max Planck Institute for Mathematics in the Sciences
3-month fellowship. Amount: €1,650 per month.
- 2026 **Pre-Doctoral Fellowship**, Max Planck Institute of Molecular Cell Biology and Genetics
3-month fellowship. Amount: €1,753 per month.
- 2018 **Scholarship for the Scuola di Orientamento Universitario**, SNS, Sant’Anna, and IUSS
Selected among the top 230 Italian students.
- 2016 **WEP Scholarship**, WEP Italy
Awarded €1,000 for academic achievement, extracurricular involvement, and personal merit.

- 2016 **ITACA Scholarship**, INPS
Awarded €7,650 for a six-month study-abroad period during the academic year. 1st place nationally.
- 2013 **XXIII Olympiad of Logical and Mathematical Games**, Gioiamathesis
2nd place.

Teaching Experience

- 2024–2025 **Teaching Assistant**
University of Trento
Mathematical Analysis 3, Department of Physics, Prof. V. Agostiniani, A.Y. 2025/2026.
Geometry and Linear Algebra, Department of Information Engineering and Computer Science, Prof. A. Bernardi, A.Y. 2025/2026, taught in English.
Geometry A, Department of Mathematics, Prof. E. L. Solá Conde, A.Y. 2024/2025.
Geometry and Linear Algebra, Department of Information Engineering and Computer Science, Prof. E. Postinghel, A.Y. 2024/2025.
Geometry and Linear Algebra, Department of Information Engineering and Computer Science, Profs. G. Occhetta and E. Postinghel, A.Y. 2024/2025, taught in English.
Mathematics and Statistics 1, Department of Biology, Profs. A. Bernardi and E. Ballico, A.Y. 2024/2025.

Talks and Posters

- Mar 2026 **Flag varieties: from minimal orbits to a dual Terracini lemma**, TensorDec Lab Seminars, University of Trento.
- Feb 2026 **Impossible ecologies via Grassmannians**, Coffee Chalk Talks, MPI-CBG.
- Feb 2026 **Secant varieties of flag varieties**, PhD Selection Week, MPI-CBG. [Poster](#).
- Feb 2026 **Tensors and their geometry**, Mathematical Structures and Applications Group Seminars, MPI-CBG. [Slides](#).
- Jul 2025 **Geometric constrained variational calculus**, Seminar for Differential Geometric Methods, University of Trento.
- Jul 2025 **Blow-up of $\mathbb{P}^2$ and minimal model program**, Seminar for Algebraic Geometry II, University of Trento.
- Oct 2024 **Noether's normalization and homotopy continuation**, Seminar for Computational Algebra, University of Trento. [Slides](#).
- Sep 2024 **The algebraic geometry of tensor decomposition**, Baby Steps Beyond the Horizon, Będlewo. [Slides](#).
- Jun 2024 **Affine toric geometry**, Seminar for Toric Geometry, University of Trento.
- Feb 2024 **Pattern formation in a model of acute inflammation**, Seminar for Mathematical Biology, University of Trento.

Conferences, Workshops, and Schools

- Oct 2025 **Masterclass: Tensors and their Applications**, University of Trento
- Aug 2025 **Scuola Matematica Interuniversitaria**, University of Perugia
Algebraic Geometry, Prof. Brian Harbourne, Grade: A+.
Complex Analysis, Prof. Chiara de Fabritiis, Grade: A.
- Jul 2025 **London Geometry and Machine Learning**, Imperial College London
Selected participant, mentored by Dr. Kelly Maggs.
- Jan 2025 **Apolarity in Varsavia**, Institute of Mathematics of the Polish Academy of Sciences
Invited by Prof. Jarosław Buczyński.
Co-authored [workshop notes](#) with O. R. Fité under the supervision of Dr. S. Canino.
- Nov 2024 **Masterclass: Tensor Decompositions in Multi-Omics**, University of Trento
- Jul 2024 **9th European Congress of Mathematics**, Seville
- Oct 2023 **Masterclass: Algebraic Vision and Multifocal Tensors**, University of Trento

Service and Outreach

- Mar 2026 **Il giro del mondo in 80 Prodotti**, Trento
Co-organized interactive booths on the history of multiplication for Pi Day.
- Oct 2025 **TensorDay 2025**, University of Trento
Co-organized with the TensorDec Laboratory.
- Apr 2025 **Fuori dall'ombra: Panchine Rosse**, University of Trento
Co-organized the *Matilda effect* section.
- Dec 2024 **TensorDay 2024**, University of Trento
Co-organized with the TensorDec Laboratory.

Skills

Languages. Italian: native. English: fluent (TOEFL iBT 111/120).

Digital skills. Macaulay2, SageMath, C, MATLAB, LaTeX, Java.